

Comparison of Event-Based Algorithms for Experimental Two-Dimensional, Two-Component Velocimetry

Sidaard Gunasekaran*^{ID}

University of Dayton, Dayton, Ohio 45469

Michael Mongin[†]

U.S. Air Force Research Laboratory, Wright–Patterson Air Force Base, Dayton 45433, Ohio
and

Osama A. AlSattam[‡] and Keigo Hirakawa[§]

University of Dayton, Dayton, Ohio 45469

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Event-based cameras, which operate without traditional frames and instead capture asynchronous changes in intensity, offer a transformative approach to flow diagnostics. This study investigates the application of event-based methods for velocimetry around an SD7003 wing and compares their performance to traditional particle image velocimetry (PIV). Four event-based algorithms [Weighted Tracking (Linear and Non-Linear fits), Kalman Filter, and Motion Compensation] are evaluated using a combination of freestream, near-surface, and wake flow scenarios. Results demonstrate that event-based methods successfully replicate global flow trends observed in PIV, including mean and median velocity estimates, while also capturing subtle flow features such as near-surface pressure gradients and wake asymmetry with higher resolution. Unlike PIV, which is constrained by frame rate and windowed processing, event-based methods track individual tracer particles, resolving flow features even in regions with steep gradients. The Weighted tracking methods exhibited the closest agreement with PIV, while Kalman Filter and Motion Compensation methods showed slight biases in U velocity, hypothesized to result from input parameter tuning. The findings highlight the potential of event-based methods as a viable candidate for flow diagnostics.

Nomenclature

a_x, b_x, c_x	=	coefficients of polynomial fit
C_{Temporal}	=	temporal contrast
c	=	wing chord
EB-PIV	=	event-based particle image velocimetry
ev_x, ev_y	=	event data location
$I(t)$	=	intensity of each pixel
n	=	number of events per track
PIV	=	particle image velocimetry
r	=	radius threshold from tracker center
T_x, T_y	=	center locations of active tracks
U, V	=	velocities in X and Y directions
U_∞	=	freestream velocity
x, y	=	coordinates in X and Y directions
$\bar{x}, \bar{y}$	=	vector locations of particle tracks
α	=	angle of attack

I. Introduction

EVENT-BASED cameras, also known as dynamic vision sensors (DVSSs) or Address-Event Representation cameras, were initially developed by Mahowald to emulate the human retina's sensitivity to motion [1]. Unlike traditional cameras, event-based cameras

operate in a frame-free manner, capturing only changes in light intensity rather than recording complete frames at regular intervals. This asynchronous output is characterized by the independent operation of each pixel, which transmits events when relative changes in intensity surpass a threshold [2]. Advances in DVS technology, pioneered at the Institute of Neuroinformatics, University of Zurich, have been extensively detailed by Lichtsteiner et al. [2] and Posch et al. [3]. The working principle of event-based cameras revolves around detecting temporal contrast changes in intensity. Each pixel generates an event when the logarithmic intensity contrast exceeds a predefined threshold, defined mathematically as

$$C_{\text{Temporal}} = \frac{1}{I(t)} \frac{dI(t)}{dt} \quad (1)$$

where $I(t)$ represents the intensity at time t . The camera outputs an asynchronous stream of events, consisting of four data columns: x and y (pixel coordinates), t (timestamp in microseconds), and p (polarity, representing on or off states). The logarithmic sensitivity of DVS sensors enables a high dynamic range (up to 120 dB) and low pixel latency (as little as 15 μs under optimal lighting) [4].

Operating with a sample clock rate near 1 MHz, these cameras achieve a temporal resolution of approximately 1 μs . Figure 1 illustrates a simulated comparison between conventional and event-based cameras for the circular motion of particles in a field of view. Conventional cameras produce frame-based outputs at fixed intervals δt , capturing static snapshots. Conversely, event cameras provide an asynchronous stream of timestamped events, exclusively recording motion without static background information.

The unique non-frame-based approach of event-based cameras positions them as promising tools for velocimetry. Table 1 highlights the comparison between traditional time-resolved particle image velocimetry (PIV) and event-based PIV. Traditional PIV relies on precise synchronization between the camera and laser as well as tailored delays between laser pulses to capture adequate particle displacement (8–10 pixels) within the sensor's field of view. It also requires homogeneous seeding for accurate velocity estimation through cross-correlation.

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*Associate Professor, Mechanical and Aerospace Engineering Department; gunasekaran1@udayton.edu. Senior Member AIAA.

[†]Aerospace Engineer; michael.mongin@us.af.mil. Member AIAA.

[‡]Ph.D. Student, Department of Electrical and Computer Engineering; alsattam1@udayton.edu.

[§]Professor, Department of Electrical and Computer Engineering; khirakawa1@udayton.edu.

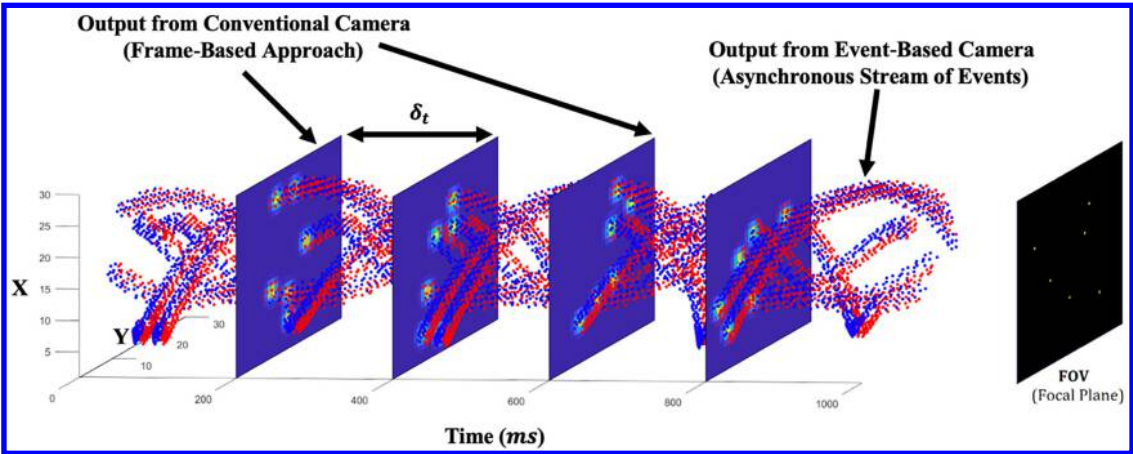


Fig. 1 Comparison of conventional frame-based and event-based camera outputs for particles undergoing circular motion [21].

Table 1 Key differences between traditional time-resolved PIV and event-based PIV

Characteristics	Metrics	Time resolved-traditional PIV	Event-based PIV
Experimentation	Synchronization	Required between camera and laser	No synchronization required
	Calibration	Requires calculating time delay and frame rate between pulses using FOV dimension	Only need FOV dimension. No time delay required.
Camera	Seeding Type	Requires uniform and homogenous seeding	Does not require uniform and homogenous seeding
	Resolution	High spatial resolution	Limited spatial resolution
	Output Type	Synchronized frames	Asynchronous events
Light source	Type	High-power double pulsed lasers for wind tunnel and continuous lasers for water tunnel	Continuous laser for wind and water tunnels
	Safety	Class IV	Class IIIb (safer) and Class IV
Analysis	—	Plethora of tools developed over decades (denoising, cross-correlation, etc.)	Algorithm development in progress. No standard algorithm exists.
Accuracy	Spatial resolution	Interrogation Windows	Individual particle tracking
	Temporal resolution	Depends on MP of camera	Up to 100 μ s and does not depend on MP of camera
System cost	—	More than \$80,000	Cost-effective (less than \$15,000)

Event-based velocimetry, on the other hand, eliminates synchronization and time-delay requirements due to its frame-free nature. Although uniform tracer particles improve tracking, they are not strictly necessary. Event cameras operate with safer Class IIIb lasers instead of the high-power Class IV lasers used in traditional methods. However, a significant limitation of event-based cameras is their resolution; the maximum currently available resolution is 1 MP (Megapixel), compared to up to 16 MP or higher for traditional PIV cameras. Furthermore, traditional PIV benefits from decades of algorithmic development for denoising and filtering, whereas event-based velocimetry is limited by the scarcity of validated algorithms for processing noisy event data. Addressing these gaps is one of the core objectives of this study.

A. Use of Event Camera for Flow Diagnostics

DVSs, commonly referred to as event cameras, have increasingly been employed to visualize and quantify fluid flows. Drazen et al. [5] demonstrated the potential of a DVS sensor for particle tracking velocimetry (PTV) in dense solid/liquid two-phase pipe flows. Similarly, Ni et al. [6] used event-based imaging to track 12 μ m microspheres, successfully capturing Brownian motion. Borer et al. [4] achieved the first application of PTV in air, employing event-based cameras to track sparsely seeded helium-filled soap bubbles within a large experimental setup featuring a 1 m field of view. Rusch and Rösgen [7,8] introduced TrackAER, a real-time event-based particle tracking system that allowed for real-time visualization of particle trajectories, making it particularly useful for large experimental setups where traditional high-speed imaging methods struggle with data bandwidth and storage constraints. Wang et al. [9] further advanced this field by implementing stereoscopic PTV with two event-based

cameras, enabling three-dimensional particle tracking in a smaller volume. More recently, Willert and Klinner [10] used event cameras to measure velocity fields in the wake of a circular cylinder in both wind and water tunnels, benchmarking their results against the traditional two-dimensional, two-component (2D2C) PIV methodology.

Beyond fluid flow applications, event cameras have also found use in various domains, including industrial automation, Internet of Things, monitoring systems, automotive and mobility technologies, and medical devices [11]. A comprehensive review of event-based sensing technology and its foundational concepts is available in Tayarani-Najaran and Schmuker [12]. Although existing algorithms for *optical flow* detection, used in applications like detecting vehicles, trees, and people, are highly effective in general-purpose scenarios, they are not optimized for fluid flow measurements. Such algorithms rely on events across broader areas, such as edges and curves, whereas fluid flow analysis involves tracer particles that, when focused properly, occupy only a few pixels. Several algorithms have been proposed that relies on optical flow methods [13–18] and neural networks [19,20] instead of cross-correlation to perform particle image and particle tracking velocimetry but only a very few is applicable to particle tracking from event cameras.

To harness the capabilities of event cameras for velocimetry, custom algorithms must be developed to filter, denoise, and extract velocity information from raw data. Some methods attempt to create virtual *frames* by time-sampling event data and processing them with traditional cross-correlation techniques. However, this approach introduces challenges, such as denoised data creating empty interrogation regions that reduce correlation quality, and windowed interrogation techniques decreasing spatial resolution.

Table 2 provides a comparative overview of key aspects such as test setups, cameras used, seeding particles, lighting, processing algorithms, and spatiotemporal resolutions across these notable studies. The algorithm developed by Drazen et al. [5], Wang et al. [9], and Willert and Klinner [10] along with other methods were compared with a synthetic dataset in a prior work [21,22].

In this paper, we evaluate and compare the performance of three in-house developed algorithms, Weighted tracking with linear fit, Weighted tracking with nonlinear fit, and Kalman filter-based tracking and the Motion Compensation algorithm developed by Willert and Klinner [10] against the traditional PIV results. Although Willert's study demonstrated the feasibility of event-based velocimetry by comparing its performance with traditional PIV, it relied on a windowed interrogation approach akin to conventional PIV. This approach, while effective, inherently limits spatial resolution and does not fully leverage the capabilities of event cameras for tracking individual particles.

In contrast, the methods employed in the current study are particle based, enabling the tracking of individual particle motion throughout the flowfield. This particle-level resolution provides a more detailed and, hypothetically, a more accurate depiction of velocity fields, avoiding the loss of spatial information associated with windowed techniques. Although this study benchmarks event-based algorithms against traditional PIV, given its widespread use, we recognize that traditional PTV, despite its advances in multiframe tracking and hybrid algorithms, remains fundamentally frame based and constrained by camera frame rates, exposure times, and synchronization requirements. By contrast, event-based methods operate asynchronously and continuously, recording pixel-level intensity changes with microsecond temporal resolution. This frame-free

approach enables high-fidelity tracking even in regions with sharp velocity gradients or near-wall features, where both PIV and traditional PTV often encounter limitations. Our prior work [5] included a comparison between event-based tracking algorithms and traditional PTV using synthetic flowfields. The current study extends this line of inquiry through experimental validation using PIV, and future work will incorporate frame-based PTV analysis.

II. Processing Algorithms Used

Four distinct algorithms were employed to estimate velocity from the event data. These algorithms, described in the following, highlight both particle-based and image-based approaches to leveraging event camera data for velocimetry. A detailed, step-by-step explanation of the in-house Weighted tracking algorithm is provided here. Descriptions of the other algorithms are summarized, with additional implementation details available in the referenced literature.

A. Weighted Tracking with Linear and Non-Linear Fit

The workflow of the Weighted tracking algorithm is shown in Fig. 2.

1. Preprocessing

The first step in the algorithm involves processing the raw data using various filters to eliminate noise. Any particle motion captured by the event camera results in two events, a positive event in the direction of the movement indicating an increase in intensity in the pixel, followed by a negative event indicating the particle leaving the pixel. Therefore, true particle trajectories will have both positive and negative polarities. Events without these polarity pairs

Table 2 Overview of experimental setups and methodologies from key event-based velocimetry studies: Drazen et al. [5], Wang et al. [9], and Willert and Klinner [10]

Reference	Drazen et al. [5]	Wang et al. [9]	Willert and Klinner [10]	
Velocimetry setup type	2C2D	3C3D	2C2D	
Test section dimensions, mm	Pipe (50 × 2700)	Hexagonal tank	Water tunnel (400 × 250 × 60)	Wind tunnel (76 × 76 × 1500)
Article of investigation	Flow in pipe driven by pump	Magnetic stirring in a tank	Flow driven by a pump	Wake of a cylinder
FOV dimensions, mm	7.75 × 7.75	6.23 × 3.6 × 5.4	54 × 30	31 × 17.5
Magnification, $\mu\text{m}/\text{pixel}$	60.55	15	42.3	41.2
Event camera used	DVS (128 × 128 pix)	Prophesee PEK3SHEM (480 × 360 pix)	Prophesee EVK2 HD (1280 × 720 pix)	
Seeding particles	Polystyrene (950 μm)	White polyethylene (90–106 μm)	Silver coated glass (10 μm)	Paraffin droplets (1 μm)
Lighting	5W CW laser	Collimated LED	5.6W CW laser	4W CW laser
Algorithm used	Particle tracker	Event-based optical flow	Motion compensation	
Spatial resolution	Per particle	Per volume	40 × 40 pixels	
Temporal resolution, ms	20	NA	20	

3D3C, Three dimensional three component.

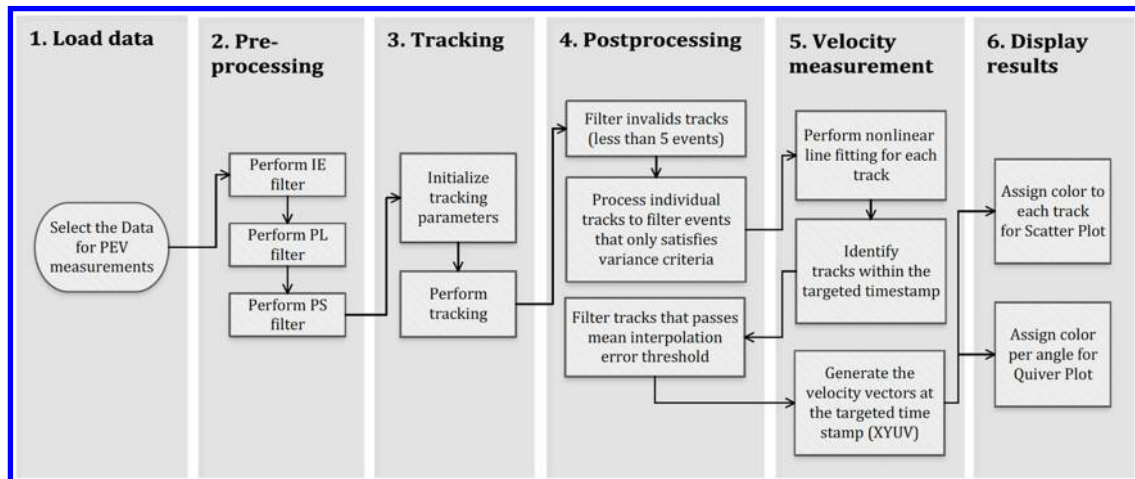


Fig. 2 Workflow developed for obtaining velocity data from event camera using Weighted tracking method (PEV, Particle Event Velocimetry).

are considered noise. The Inceptive Event (IE) filter, developed by Baldwin et al. [23] at the University of Dayton, removes this noise. Willert [24] proposed using a pulsed laser instead of a continuous one to tackle this issue. However, this approach negates the camera's microsecond resolution advantage. After eliminating redundant data using the IE filter, Pulse Filter (PL) is applied to validate the polarity pairs, ensuring that a positive event is followed by a negative event. This Polarity Filter (PF) is designed to identify a pair of positive and negative events within a certain threshold. Because the particle tracks are much cleaner after denoising, the Positive Event (PS) filter is used to select only events with positive polarity, which are then used for tracking in the subsequent step.

B. Particle Tracking

Tracking is performed using the Weighted tracker method. The cost function for particle tracking is given by Eq. (2),

$$W = n * \left[\frac{(ev_x - T_x)^2 + (ev_y - T_y)^2}{r^2} - 1 \right] \quad (2)$$

where (T_x, T_y) represents the center of all active tracks and (ev_x, ev_y) is the location of the event under investigation; n is the number of events for each track, and r is the radius threshold from the tracker center.

If the calculated weight is negative, the new event is within the assigned threshold radius r and the number of events in the track is updated. If W is positive, then the event is not sufficiently close to any tracks based on the assigned threshold r . Figure 3 show the Weighted trackers handle the events, assigning each event to a track and updating its information with every cycle. The intensity of the red color corresponds to the weight of the track determined by Eq. (2). Tracks 3 and 4 have more weight than the other tracks.

C. Postprocessing

Because the Weighted tracker is based on the (x, y) positions of the events, it might inadvertently include events from a neighboring

track within the current track. To resolve this issue, the following three posttracking filters are applied:

Threshold Filter: For the threshold filter, particle tracks with fewer than a certain number of events (e.g., fewer than five events) are eliminated.

Mean Squared Line Fitting Error Filter: The mean squared line fitting error filter determines the difference between the data points and a line fitted through the particle tracks. If the difference exceeds a certain threshold, it is deemed not valid and will be removed. A schematic of this filter is shown in Fig. 4.

Pruned Exact Linear Time Filter: For the Pruned Exact Linear Time Filter, here is a possibility that two tracks may overlap at different times. In such cases, both the Threshold and Mean Squared Line Fitting Error filters might fail to detect the corresponding location and direction of the track. To address this, a Pruned Exact Linear Time (PELT) filter [25] is employed. This filter is based on the inherent assumption that if events on the same track belong to the same particle; then, the timestamps of those events will follow a linear pattern. Therefore, a sudden change in the timestamps of events along a particle track can be identified and the affected portion of the track can be removed.

D. Velocity Estimation

The velocity can be extracted from the particle tracks using both linear and nonlinear fitting methods. Because the vector locations of particle tracks, $\bar{x}$ and $\bar{y}$ are known, the coefficients of the fit can be obtained by

$$A_x = (M^T M)^{-1} M^T \bar{x} \quad (3)$$

where

$$M_{\text{Non Linear}} = \begin{bmatrix} t_1^2 & t_1 & 1 \\ \vdots & \vdots & \vdots \\ t_n^2 & t_n & 1 \end{bmatrix} \quad M_{\text{Linear}} = \begin{bmatrix} t_1 & 1 \\ \vdots & \vdots \\ t_n & 1 \end{bmatrix} \quad (4)$$

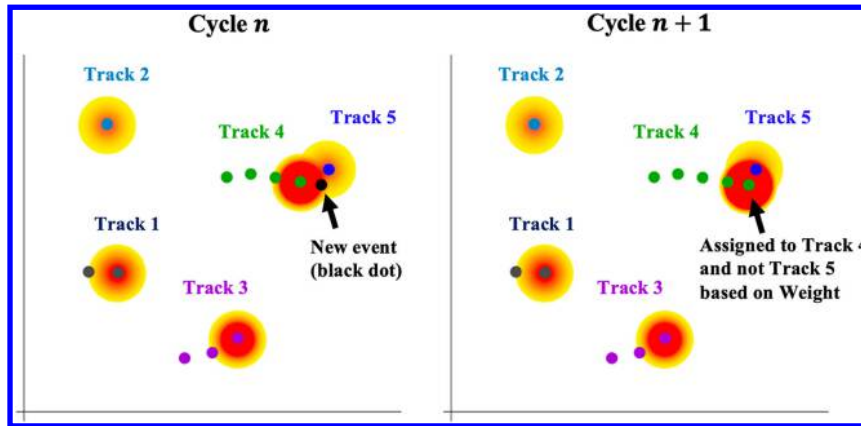


Fig. 3 Demonstration of Weighted tracking: a new event is assigned to the track with the highest weight rather than the nearest track.

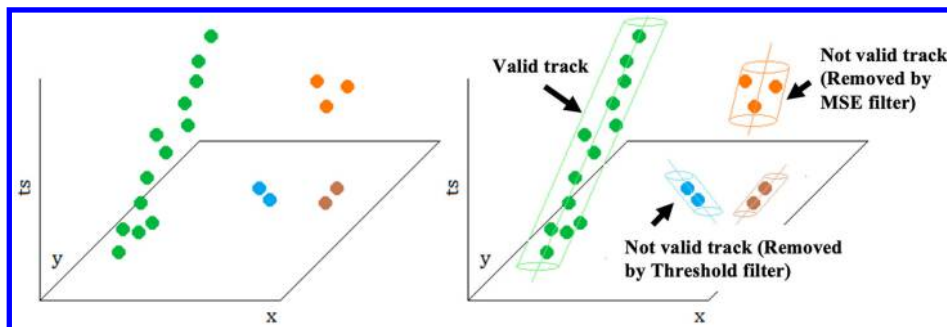


Fig. 4 Schematic showing the mean squared line fitting error filter (MSE, mean square error).

The coefficients of the fit is

$$A_x = \begin{bmatrix} a_x \\ b_x \\ c_x \end{bmatrix} \quad (5)$$

Once A_x and A_y are known, the velocity and acceleration of the particles can be determined by

$$V_x = \frac{\partial}{\partial t} (a_x t^2 + b_x t + c_x) = 2a_x t + b_x$$

$$Ac_x = \frac{\partial^2}{\partial t^2} (a_x t^2 + b_x t + c_x) = 2a_x \quad (6)$$

The same operations can be performed to determine V_y and Ac_y . The differences between the linear and nonlinear interpolation methods for obtaining velocity are shown in Fig. 5. Although the linear method only displays the velocity, the nonlinear method reveals curved trajectories, which indicate the acceleration of the particle. Traditional velocimetry techniques are unable to provide such information, highlighting another advantage of event-based velocimetry.

The results from overall workflow of the event-based velocimetry algorithm developed at the University of Dayton is shown in Fig. 6. The raw data, which are 100% of the data obtained directly from the event camera, are run through the various filtering, denoising, and tracking process as mentioned previously. At the end of the processing algorithm, only 15% of the raw data is used to create the velocity

plot. This underscores the need of estimating the overall errors and the accuracy of the velocity information, as substantial denoising and filtering are employed at various stages of data processing.

E. Kalman Filter-Based Tracking

The Kalman filter-based tracking algorithm, described in detail by AlSattam et al. [22,26], employs a causal filtering approach to estimate particle positions and velocities while accounting for noise and uncertainties in the measurements. This method recursively predicts the state of each particle and updates the predictions using incoming event data. By applying a weighted average of the prediction and the observation, the Kalman filter minimizes the impact of measurement noise and erratic particle behavior. This probabilistic approach is particularly advantageous in highly dynamic environments, such as turbulent fluid flows, where rapid fluctuations and incomplete observations present significant challenges for traditional tracking algorithms. The Kalman filter's robustness makes it a powerful tool for real-time applications requiring noise-resilient velocity estimation.

F. Motion Compensation Algorithm

The Motion Compensation algorithm, originally introduced by Gallego et al. [27] was adapted and implemented by Willert and Klinner [10] for event-based velocimetry applications. Unlike particle-based tracking methods, this technique adjusts the observation frame to align with the expected flowfield. The algorithm predicts velocities by iteratively realigning the frame and counting the number of events in each alignment, enhancing the accuracy of particle tracking. This approach is particularly effective in

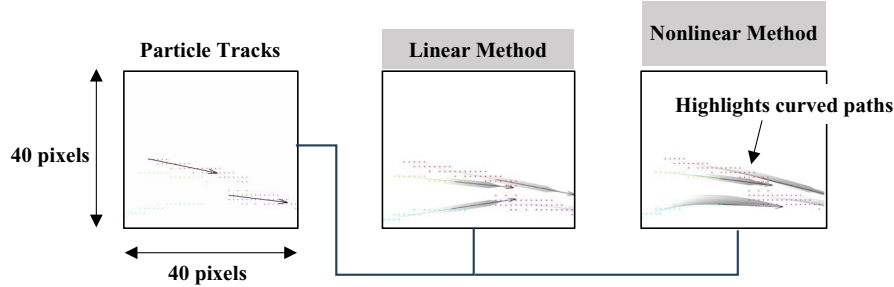


Fig. 5 Schematic showing the differences between the linear and nonlinear velocity estimation methods.

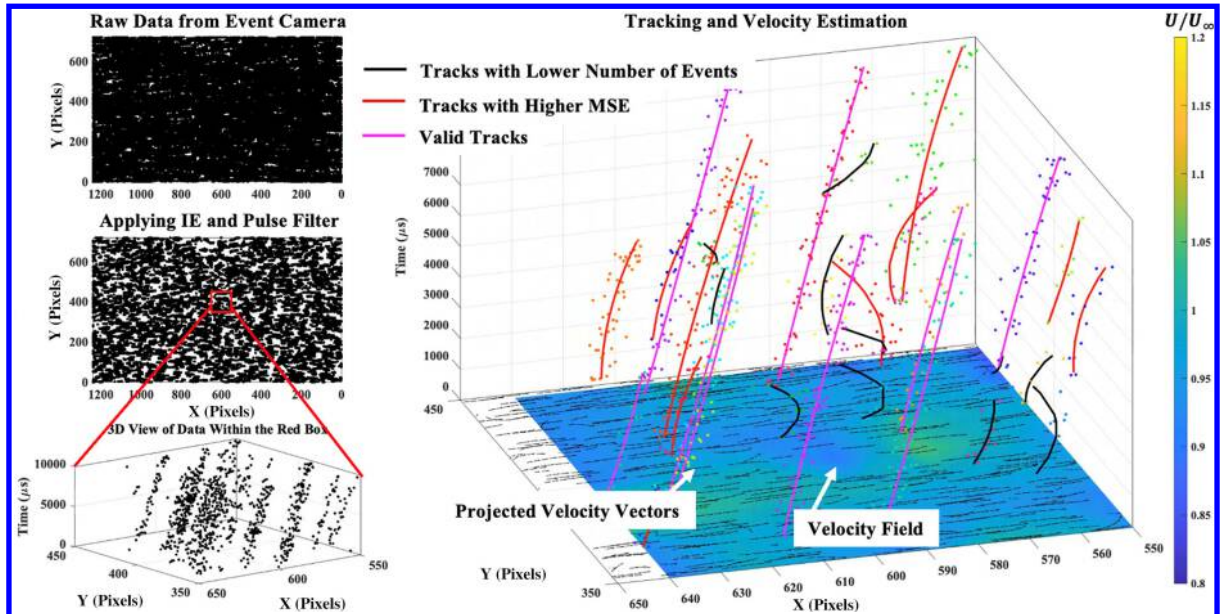


Fig. 6 Results at every stage of the event-based algorithm developed in-house at the University of Dayton.

high-speed or directional flows, where background interference or motion artifacts would otherwise hinder traditional tracking methods. However, as a windowed approach, it sacrifices particle-level resolution and is less adept at resolving individual particle motions compared to particle-based techniques.

III. Experimental Setup

A. University of Dayton Water Tunnel

The event-based velocimetry and traditional velocimetry experiments were conducted at the University of Dayton Water Tunnel, shown in Fig. 7. This facility features a horizontal free-surface water tunnel with a 6:1 contraction ratio.

The test section measures 38 cm in width, 46 cm in height, and 152 cm in length, with a freestream speed range of 7.5 to 38 cm/s. The tunnel is equipped with a one-degree-of-freedom motion stage, comprising an H2W Technologies linear motor driven by a Xenus XTL-230-18 motor driver from Copley Controls. A rack-and-pinion gear system converts the motor's linear motion into rotational motion, enabling precise pitching of a test article within the tunnel's test section. Closed-loop feedback and control for adjusting the angle of attack and coefficient of lift are facilitated by a dSPACE MicroLabBox with BNC-type terminations, programmed using the MATLAB® LabVIEW Real-Time Interface.

B. Test Model

To visualize both the upper and lower surfaces of the wing, an SD7003 wing model was fabricated using clear resin. After three iterations with different resin materials, the final model was fabricated

using Accura ClearVue High Resolution resin, which included a clear coat as shown in Fig. 8. This material was chosen because it eliminated the need for sanding or external polishing, resulting in a smooth and transparent surface. The wing model had a span of 7.25 in., a chord of 3 in., and an aspect ratio close to 5. When immersed in the water tunnel, the model's transparency allowed the laser to pass through and illuminate both the upper and lower surfaces effectively, facilitating visualization during the experiments.

C. Test Setup for Event-Camera and Traditional Velocimetry

The wing model was mounted to a force sensor connected to a linear actuator, with the field of view positioned at the tunnel centerline to align with the model's midspan. A 45 deg mirror was placed in front of the cameras, parallel to the bottom surface of the tunnel, as shown in Fig. 9. For traditional PIV, a PCO DiMax S4 high-speed camera equipped with a 2016 × 2016 pixel CMOS (Complimentary Metal Oxide Conductor) sensor was employed. For a tracer, 60 μm Polyamide particles were used. The seeding strategy targeted a particle image density of approximately 8–10 particles per 32 × 32 pixel interrogation window in the PIV images. Although exact particle counts per window were difficult to verify across the entire field of view, seeding adequacy was manually assessed during image acquisition to ensure consistent particle distribution.

The PIV system used a 4 MP PCO DiMax camera (2016 × 2016 pixels), while the event-based system employed a 1 MP Prophesee EVK2 sensor (1280 × 720 pixels). Despite the lower resolution of the event-based camera, the fields of view differed between the two systems: the PIV setup captured approximately 90 mm × 90 mm, while the event-based system covered a larger



Fig. 7 University of Dayton Water Tunnel facility (F/T, Force/Torque).

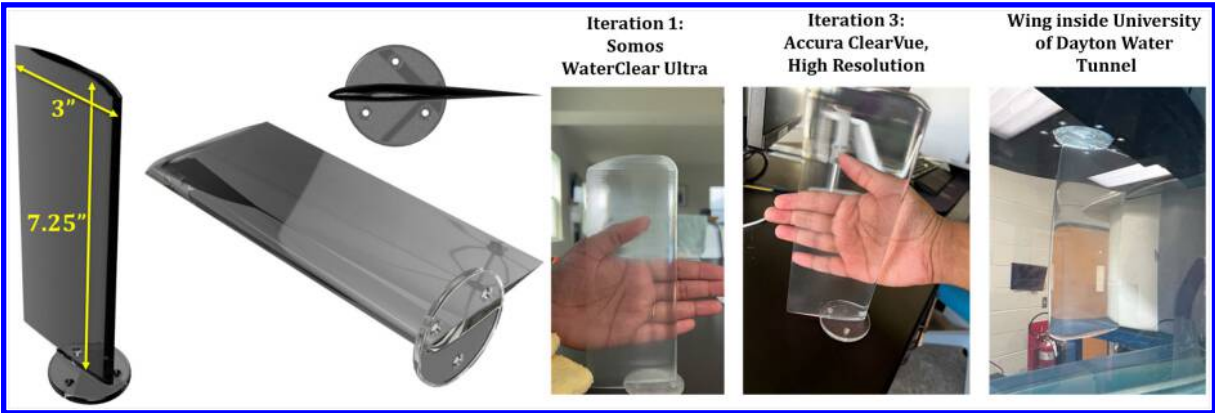


Fig. 8 Images showing the design, fabrication, and installation of the SD7003 wing model.

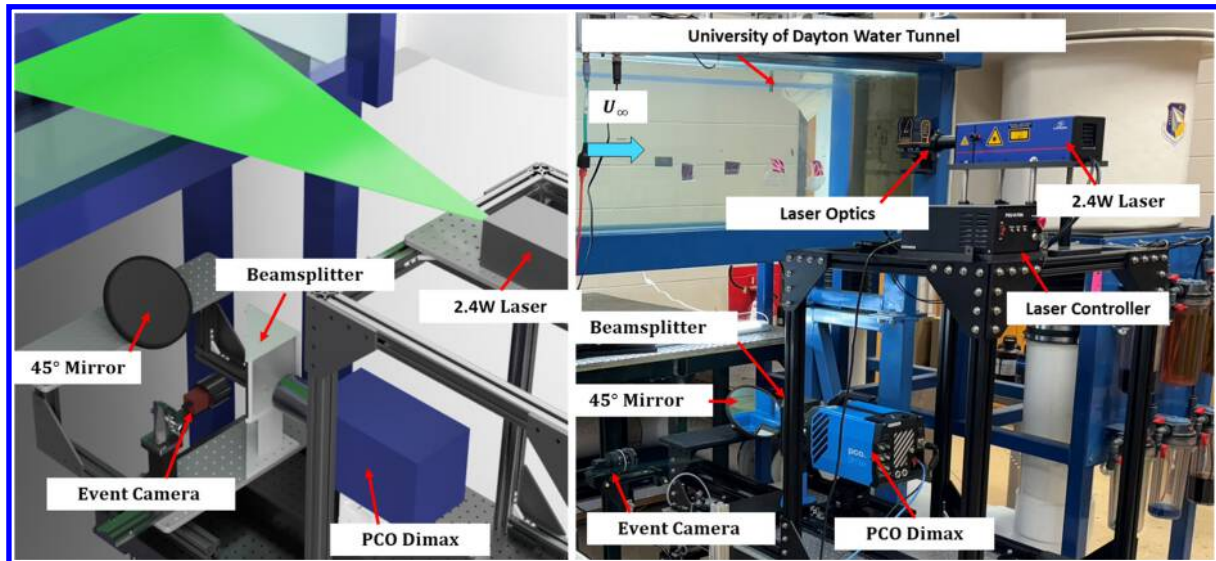


Fig. 9 Schematic and setup of event-based and traditional velocimetry systems in the University of Data Water Tunnel using dual cameras and shared optics.

area of $162 \text{ mm} \times 90 \text{ mm}$. As a result, particle density per pixel and per unit area varied slightly between systems. Nonetheless, consistent seeding conditions were maintained during simultaneous acquisition through a 50–50 beam splitter.

A representative raw image from the event-based camera is shown in Fig. 10, demonstrating individual particle traces and seeding density during one of the acquisition runs. This image illustrates the consistent event trails generated across the field of view, enabling reliable tracking by the event-based algorithms.

The event camera was calibrated using a dot plate. Because the event camera does not display an image unless motion is detected, slight vibrations were introduced to the test setup to generate detectable events. The resulting circles on the dot plate were captured over a duration of 4 s. Events were grouped into 10 ms intervals, and a circle-finding algorithm was applied to identify the field of view (FOV) for the entire 4 s data set, based on each 10 ms interval. The dot plate calibration and detected circles are illustrated in Fig. 11a, while an interquartile range (IQR) plot of the X and Y FOV obtained from the calibration algorithm is shown in Fig. 11b. The X FOV of the event camera was measured to be 162 mm, and the Y FOV was 90 mm.

The traditional PIV analysis was performed using the Fluere software package [28]. A two-pass cross-correlation approach was used, with an initial interrogation window size of 64×64 pixels,

followed by a second pass using 32×32 pixels. Both passes employed a 50% overlap between interrogation regions.

D. Test Matrix

The test matrix for the force-based experiments and simultaneous PIV and event-based velocimetry is summarized in Table 3. Force-based testing was conducted at three tunnel speeds, with angles of attack ranging from -20 to 20 deg in 1 deg increments. The Reynolds numbers for these tests ranged from 17,000 to 32,000, based on the wing chord. For traditional and event-based velocimetry, testing was performed over an angle of attack range of 0 to 10 deg, in 2 deg increments.

The parameters for traditional PIV, including the number of frames, frame rates, and exposure times, are provided in Table 3. For event-based velocimetry, 4 s of data were captured for each case, with two different high-pass filter (HPF) bias settings applied during data acquisition. The HPF bias setting determines the minimum rate of illumination change required for a pixel to output an event, allowing for filtering of slower changes in lighting. This filtering effectively removes static background elements, isolating fast-moving objects and reducing background noise. While the results presented in this paper are based on an HPF bias of 80, the sensitivity of results to other bias settings (e.g., 120) will be explored in a follow-up study. Additionally, freestream data were

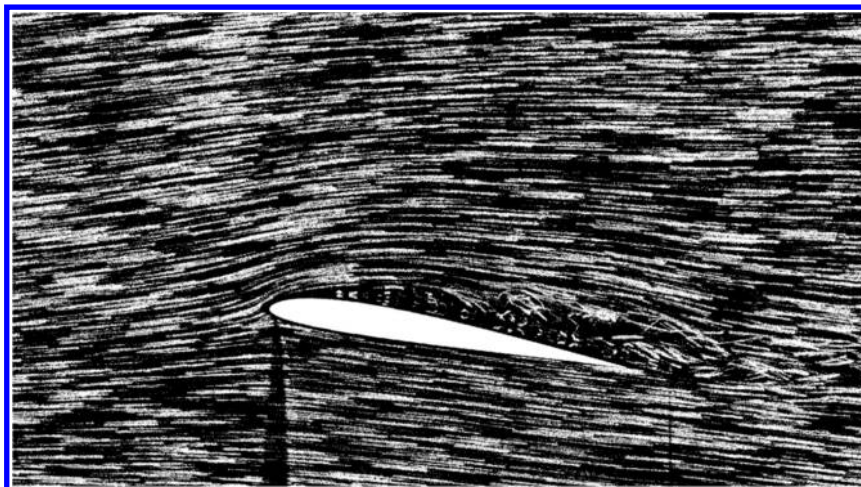


Fig. 10 Raw data from event camera of flow around the SD7003 wing.

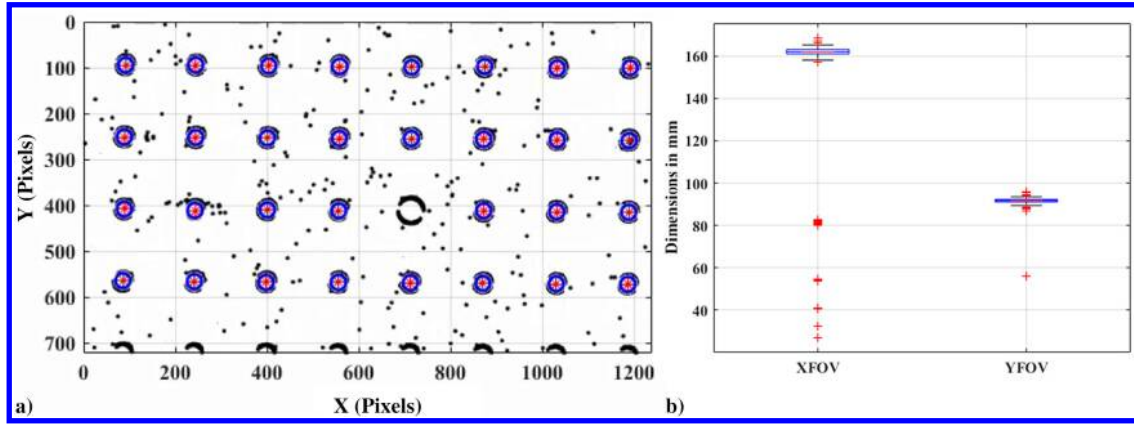


Fig. 11 a) detected calibration circles over 10 ms and b) IQR plot of X and Y FOV dimensions from 4 s of calibration data.

Table 3 Summary of test conditions, including tunnel speeds, α , Re , and camera settings for both traditional PIV and event-based velocimetry

Velocity, m/s	Force-based testing		Traditional PIV				Event-based PIV	
	AOA, deg	Re	AOA, deg	No. of time-resolved PIV frames	Frame rate, Hz	Exposure, μ s	Duration of event acquisition, s	HPF bias setting
0.2	-20 to 20	17,745	0,2,4,6,8,10	1500	238	1300	4	80, 120
0.3		26,617			330	1200		
0.37		32,827			440	800		
0.2	No-wing (freestream only)	—	—	1500	238	1300	4	80, 120
0.3					330	1200		
0.37					440	800		

collected at various speeds using both methods to establish baseline comparisons. The experimental design ensures robust velocity measurements across a range of flow conditions, enabling a detailed evaluation of both traditional and event-based velocimetry techniques.

IV. Results

A. Force-Based Results

The coefficient of lift C_L as a function of angle of attack (AOA) for the wing at three different Reynolds numbers is shown in Fig. 12. At the highest velocity tested (0.37 m/s), the lift curve slope closely follows the predicted trend from the low Reynolds number corrected lift curve slope, represented by the dashed line. Similarly, the results at 0.3 m/s exhibit a comparable trend, except for deviations in the stall angle. At 0.2 m/s, however, the lift curve slope changes noticeably around a 5 deg AOA, which is hypothesized to result from the formation of a laminar separation bubble (LSB). At higher velocities (0.3 and 0.37 m/s), the LSB is believed to break earlier, leading to a higher standard deviation in the lift coefficient between AOA of 4 and 8 deg. In contrast, at 0.2 m/s, the standard deviation in this range is lower because the separation bubble remains attached and sheds only at higher angles of attack. Based on the observed lift curve behavior at different velocities, the angle of attack range for traditional and event-based PIV measurements was narrowed to 0 to 10 deg in 2 deg increments to focus on regions of interest while minimizing potential data inconsistencies.

B. Velocimetry Results

1. Freestream

The comparison between the four event-based algorithms and the traditional PIV method is presented in this section, beginning with the freestream velocities. Figure 13 shows the normalized U-velocity contours for freestream velocities of 0.2, 0.3, and 0.37 m/s, obtained using the conventional PIV method and the event-based algorithms. The contours from traditional PIV exhibit smooth variations across

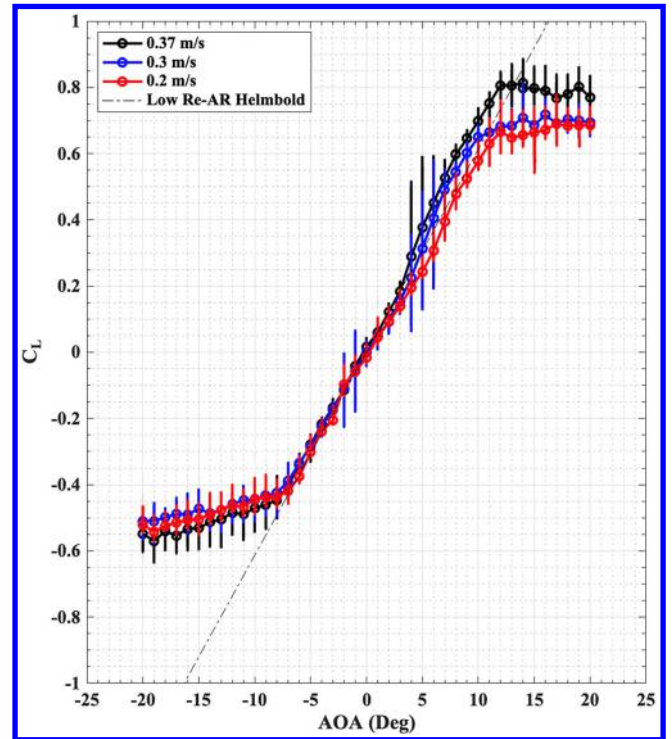


Fig. 12 Lift curves at Reynolds numbers 17,745 (0.2 m/s), 26,617 (0.3 m/s), and 32,827 (0.37 m/s); dashed line shows corrected slope.

the field of view, owing to the windowed approach with 50% overlap, which smooths gradients but reduces spatial resolution. In contrast, the event-based methods display speckled patterns across the field of view, a characteristic of particle tracking approaches that resolve individual particle trajectories. This pattern is due to the inherent

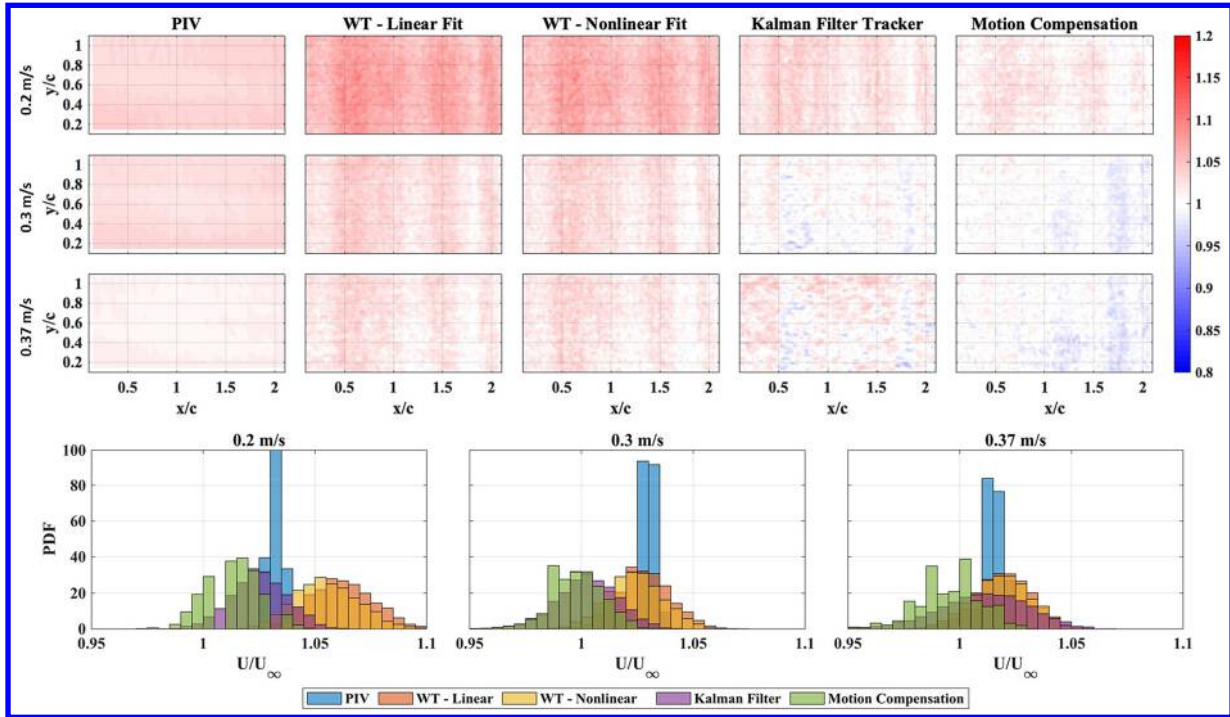


Fig. 13 (Top) U-velocity contours from PIV and event-based methods and (bottom) velocity histograms showing broader distributions and reduced bias at higher speeds.

increase in the spatial resolution when compared to PIV but introduces a less smooth representation of the flowfield. For reference, processing PIV data with smaller interrogation windows (e.g., 32×32 in the first iteration and 16×16 in the second iteration) would produce similar speckling but at the cost of reduced overall accuracy.

Both the linear and nonlinear fit methods in the Weighted tracking approach show similar behavior. However, a consistent bias is observed across the field of view for all velocities when using the Kalman filter and motion compensation methods. This bias is likely due to improper tuning of these algorithms for the specific dataset. A sensitivity study to address this will be detailed in a future paper. The velocity histograms for all methods at the three freestream velocities are also shown in Fig. 13. The PIV method produces histograms with narrower distributions and higher kurtosis, indicating less variability. In contrast, the event-based methods capture broader velocity distributions for all tested velocities. A clear bias in the mean velocity is observed for all event-based methods compared to traditional PIV, though this bias diminishes as velocity increases. For the 0.3 and 0.37 m/s cases, the Weighted tracking methods demonstrate good agreement with PIV in capturing the mean velocity. However, the Kalman filter and motion compensation methods exhibit a higher degree of bias compared to the Weighted tracking methods.

Figure 14 evaluates the statistical performance of PIV and four event-based algorithms across various freestream velocities. The mean and median generally decrease with velocity, with PIV, Weighted tracking (WT)-Linear, and WT-Non-Linear displaying similar, stable trends. However, Motion Compensation consistently underestimates these central tendencies, while the Kalman Filter exhibits significant variability and pronounced changes in both metrics. Skewness results indicate increasingly negative skew (left-tailed distributions) as velocity increases, with Kalman Filter and Motion Compensation showing the greatest sensitivity to these changes. Kurtosis reveals that the Kalman Filter produces sharp peaks and heavy tails, particularly at intermediate velocities, highlighting its susceptibility to extreme outliers. In contrast, PIV and WT methods exhibit kurtosis values slightly below 3, indicating distributions with fewer extremes and reduced tail heaviness. Overall, WT methods exhibit statistical robustness and reliability comparable to PIV, while Kalman Filter is highly sensitive to velocity

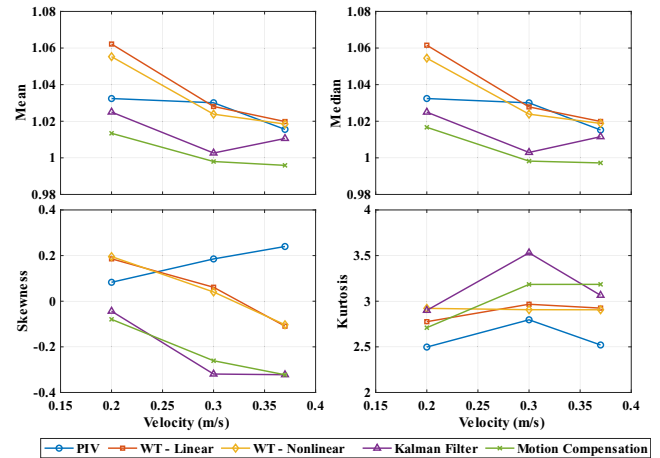


Fig. 14 Comparison of mean, median, skewness, and kurtosis; WT methods match PIV trends, while Kalman and Motion Compensation show greater skewness and variability.

changes, and Motion Compensation underestimates velocities and displays asymmetric distributions.

2. Flow Around SD7003 Wing

A comparative analysis of the normalized U-velocity contours around the SD7003 wing was conducted between the PIV method and the various event-based algorithms. The normalized U-velocity contours at different angles of attack α are shown in Fig. 15.

The rows represent angles of attack ranging from $\alpha = 0^\circ$ to $\alpha = 10^\circ$, while the columns correspond to the different algorithms used. As seen in the freestream case, the traditional PIV method produces smooth and uniform gradients around the wing, capturing higher velocities on the upper surface, lower velocities on the lower surface, and flow separation regions. The event-based methods display similar trends, with the Weighted tracking methods (WT-Linear and WT-Non-Linear) closely resembling the PIV results. Flow separation and stall behavior, especially at $\alpha = 10^\circ$ are well

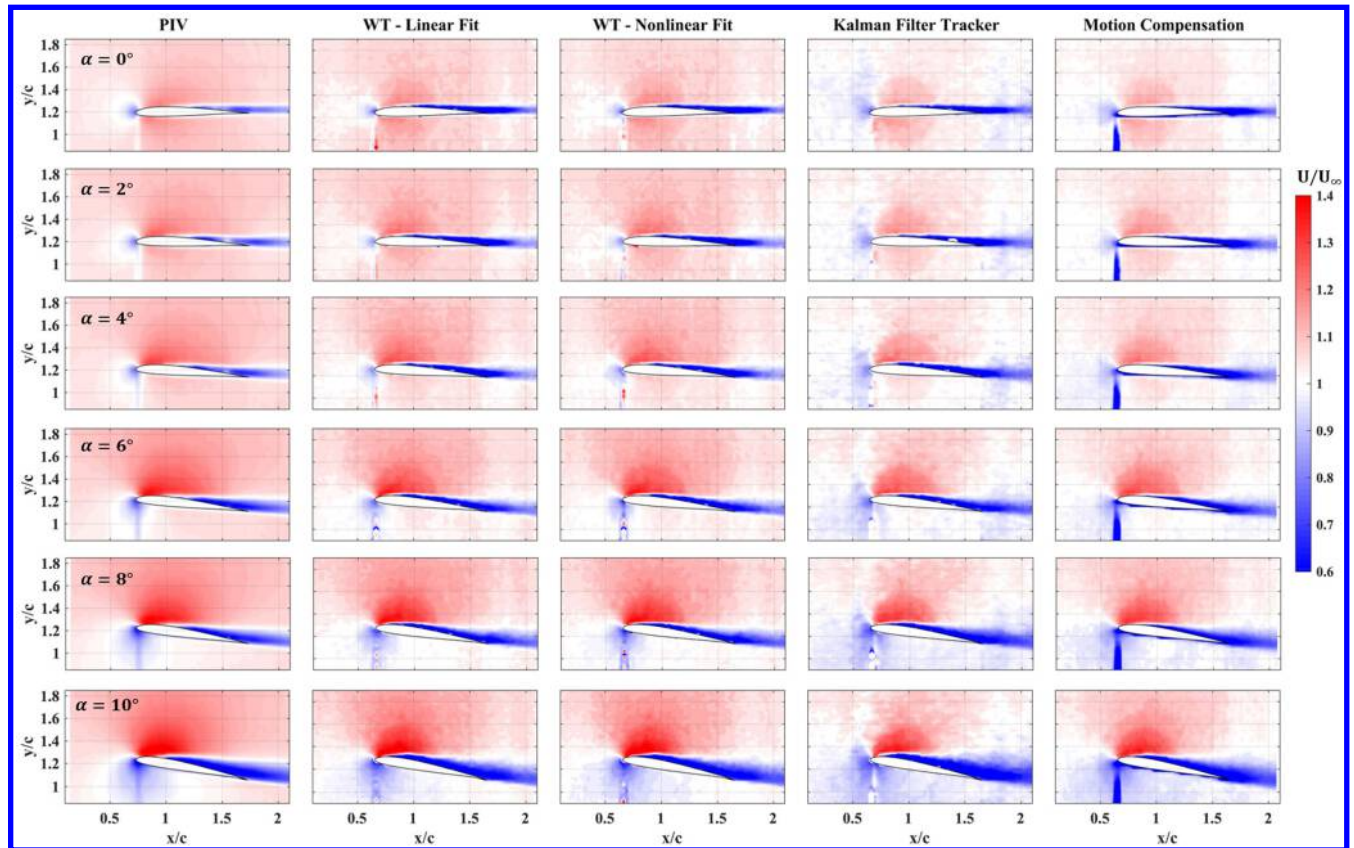


Fig. 15 Comparison of normalized U-velocity contours from PIV and event-based methods; WT methods closely match PIV, while Kalman and Motion Compensation show slight biases.

captured by both the PIV and the event-based methods. However, the Weighted tracking methods (linear and nonlinear fits) show the closest agreement with PIV, particularly in representing the transition from attached flow to separation. This consistency across angles of attack suggests that Weighted tracking methods are robust and suitable for analyzing aerodynamic behavior. The Kalman Filter and Motion Compensation methods, while capturing the general trends, show more sensitivity to velocity changes. These methods also show a slight bias in the magnitude of the contours. This bias could be attributed to the algorithmic assumptions, noise modeling, and so on that require further tuning for the flow conditions studied. This tuning, along with a sensitivity analysis, is currently under investigation and will be addressed in a future publication. Additionally, the shadow cast by the leading edge of the wing is distinctly visible in the Motion Compensation results, whereas it is less

pronounced in the other methods. Overall, at each angle of attack, the trends observed in PIV are successfully captured by the event-based methods, with the Weighted tracking algorithms demonstrating the closest alignment.

Although Fig. 15 illustrates the changes in the flowfield as a function of angle of attack, Fig. 16 depicts the variation of normalized U-velocity at different freestream velocities around the wing at a fixed angle of attack ($\alpha = 4^\circ$). At a freestream velocity of 0.2 m/s, the wake is noticeably thicker, with flow separation occurring near the quarter chord of the wing. As the velocity increases, the wake becomes thinner, reflecting reduced flow separation and enhanced boundary-layer attachment. At 0.37 m/s, the displacement thickness over the wing, as predicted by XFLR5, is shown as a green line. The experimental results, including both PIV and event-based methods, align well with the XFLR5 prediction. The event-based methods

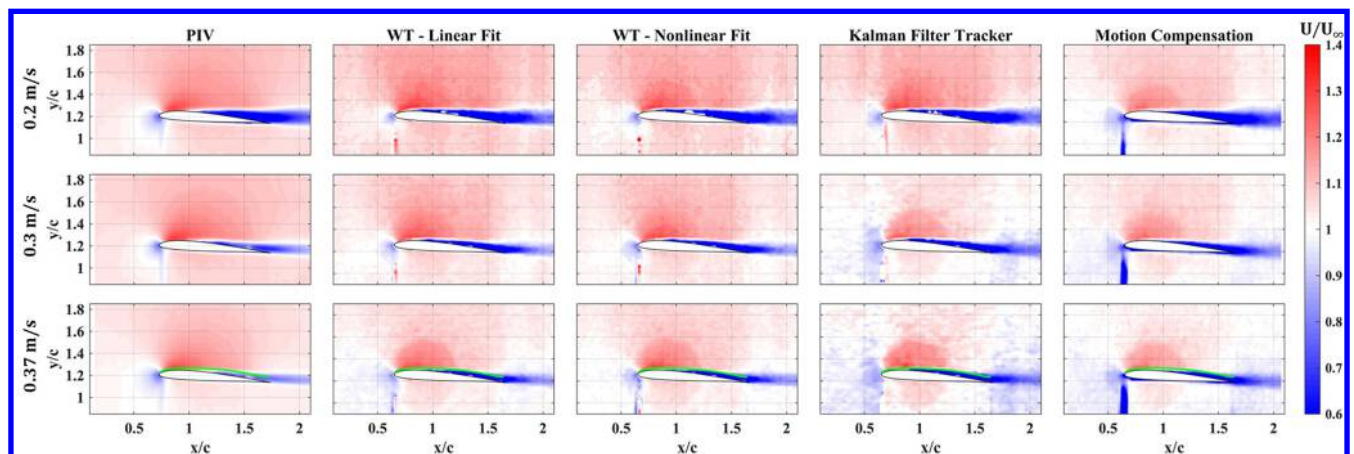


Fig. 16 Comparison of normalized U-velocity contours from PIV and event-based methods at 0.2, 0.3, and 0.37 m/s with fixed $\alpha = 4^\circ$.

once again exhibit remarkable similarity in trends and magnitudes compared to PIV. However, a consistent bias in velocity magnitude is observed in the Kalman Filter and Motion Compensation methods, which may require further refinement and parameter tuning.

In Fig. 16, the event-based methods estimate slightly lower wake velocities compared to PIV. This is further illustrated in Fig. 17, which presents freestream-normalized momentum deficit profiles at $\alpha = 0^\circ$ and $\alpha = 2^\circ$, for the three velocities tested. To account for the observed bias in the Kalman Filter and Motion Compensation methods, the wake velocities were normalized with the respective freestream velocity observed in each method, enabling a more effective comparison. As shown in Fig. 17, PIV consistently predicts lower momentum deficits in the wake region across all freestream velocities. This discrepancy can be partially attributed to a limitation in traditional PIV methods: the frame interval Δt or frame rate must be tuned for specific regions of interest in the field of view, particularly when large velocity gradients are present. In this study, Δt was optimized for the freestream velocity, ensuring tracer particles traveled 8–10 pixels per displacement in the freestream region. However, this tuning does not account for the lower particle displacements in the wake region, violating the established rule of thumb for cross-correlation accuracy. Consequently, lower cross-correlation peak magnitudes and reduced velocity estimation accuracy are expected in the wake. Event-based methods, in contrast, are not constrained by a fixed frame rate. They effectively track particles regardless of velocity variations across the field of view, enabling accurate measurements even in regions with high velocity gradients. This capability of event-based methods to excel in areas with steep gradients has been demonstrated using synthetic data in the work by AlSattam et al. [21,22].

Traditional PIV encounters significant challenges in resolving flow features near the surface of test models due to reflections caused by high-power lasers. Addressing this often requires the application of rhodamine paint to the model surface and the use of lens filters on the PIV camera. However, even with these adjustments, accurately capturing surface flow features remains difficult. Event-based methods, by contrast, are not impacted by static reflections from the test model, as the event camera only registers dynamic changes within its field of view. This enables event-based methods to better capture near-surface flow features compared to traditional PIV.

Figure 18 illustrates this advantage. The normalized U-velocity contours provide a close-up comparison between traditional PIV and the Weighted tracking method with a nonlinear fit. The displacement thickness predicted by XFLR5 is overlaid on the contours.

Traditional PIV shows a region of flow acceleration beneath the predicted displacement thickness, while the event-based method more accurately captures flow deceleration in this region. The advantage of event-based methods is further demonstrated by the coefficient of pressure C_p profiles at different freestream velocities with an angle of attack of 4 deg. Compared to C_p predictions from XFLR5, the Weighted tracking method provides a closer match than traditional PIV. Traditional PIV struggles to resolve the pressure gradients caused by the laminar separation bubble near the surface, while event-based methods capture these gradients with comparatively greater accuracy. This highlights the potential of event-based methods for resolving critical near-surface flow features that are difficult to measure using traditional PIV techniques.

To further validate the normalized U-velocity estimates from event-based methods against traditional PIV, histograms of U-velocity at various angles of attack α are shown in Fig. 19 for a freestream velocity of 0.37 m/s. Similar to the freestream results, a slight bias is observed in the event-based methods compared to traditional PIV. At low angles of attack (0 and 2 deg), the histograms are relatively narrow and symmetric, indicating stable flow. However, as the angle of attack increases, the histograms broaden significantly, reflecting increased flow variability due to separation and stall effects. The Weighted tracking methods (WT-Linear and WT-Non-Linear) closely align with PIV across all angles of attack, particularly in capturing the widening distributions at higher α . In contrast, the Kalman Filter and Motion Compensation methods exhibit greater bias, especially at higher angles, likely due to challenges in resolving separated flow regions. However, as mentioned before, these methods might likely need to be tuned further.

Figure 20 presents the variation of the mean and median of normalized U-velocity as a function of angle of attack α for all five methods at three different freestream velocities. These metrics capture the combined effects of flow acceleration over the upper surface, deceleration along the lower surface, and wake dynamics, as illustrated in Figs. 15 and 16. For all methods, the mean and median remain roughly constant with increasing α , indicating that

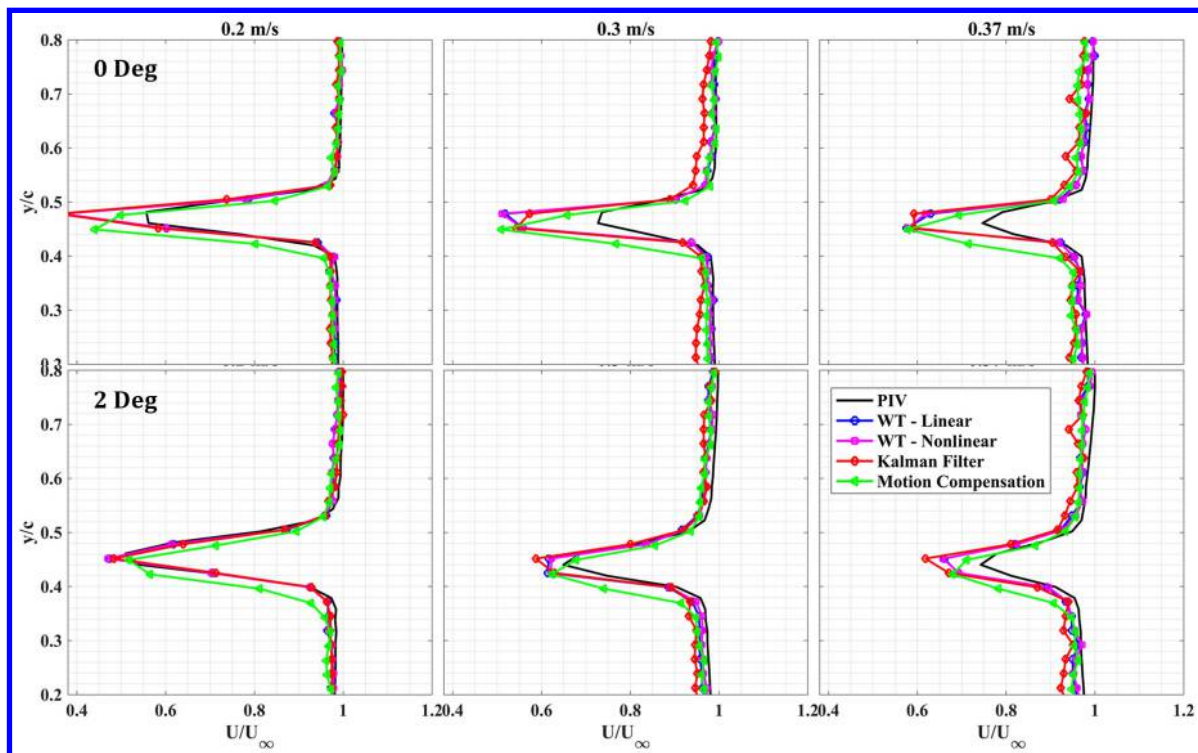


Fig. 17 Momentum deficit profiles from PIV and event-based methods at 0.2, 0.3, and 0.37 m/s for $\alpha = 0$ and 2 deg.

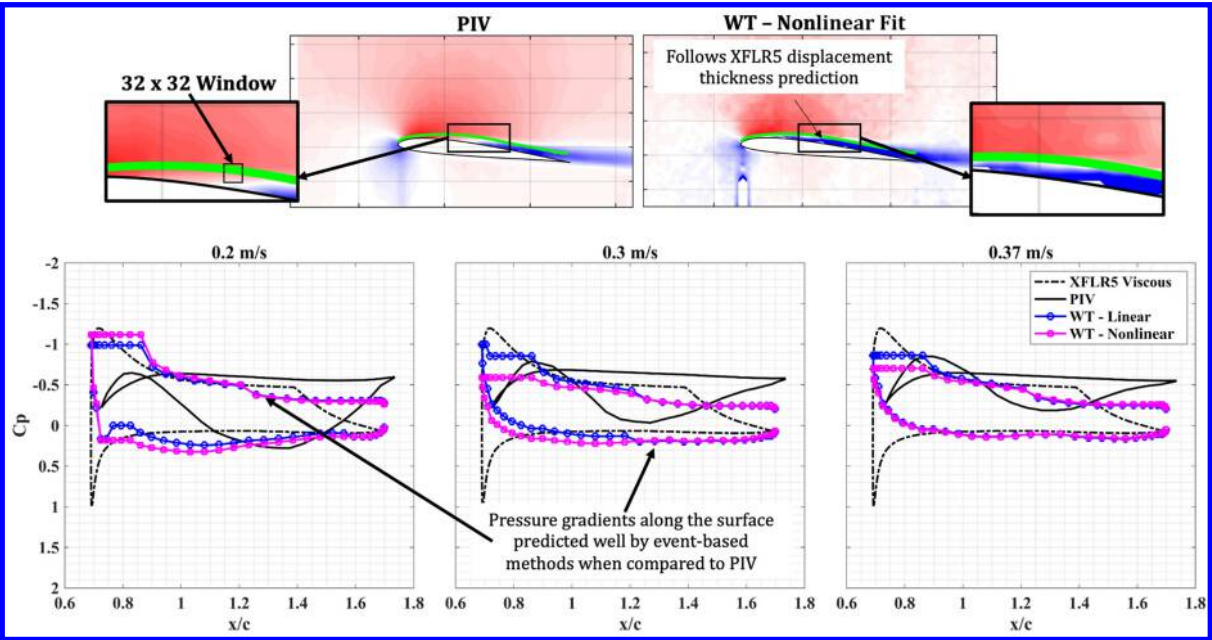


Fig. 18 (Top) U-velocity contours with XFLR5 displacement thickness and (bottom) C_p profiles showing better event-based agreement with XFLR5 than PIV.

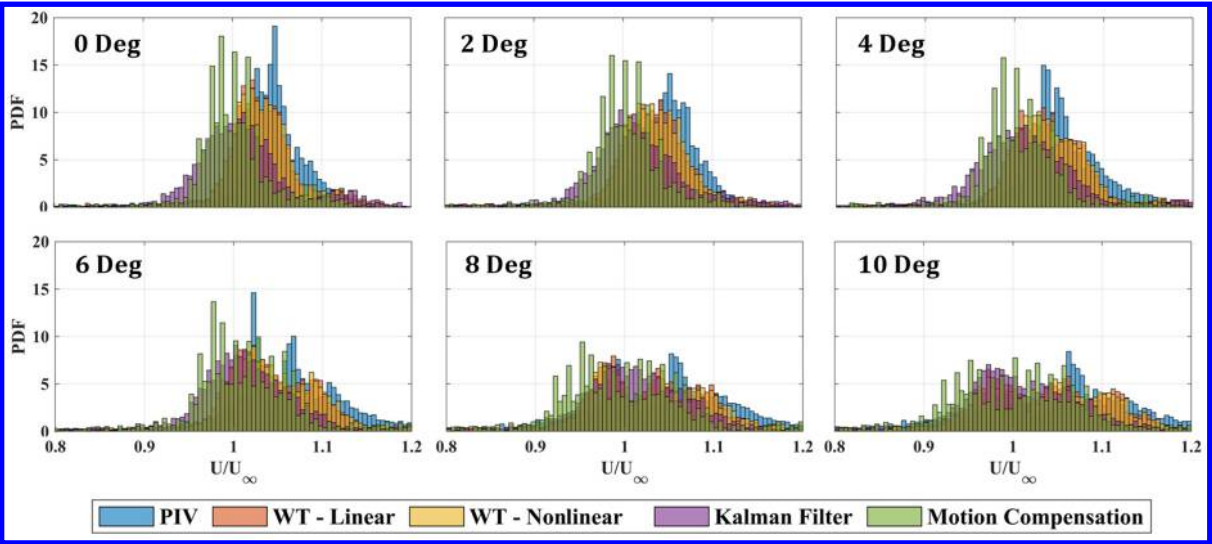


Fig. 19 Histograms of normalized U-velocity at $\alpha = 0$ to 10 deg and 0.37 m/s; event-based methods capture trends with slight bias relative to PIV.

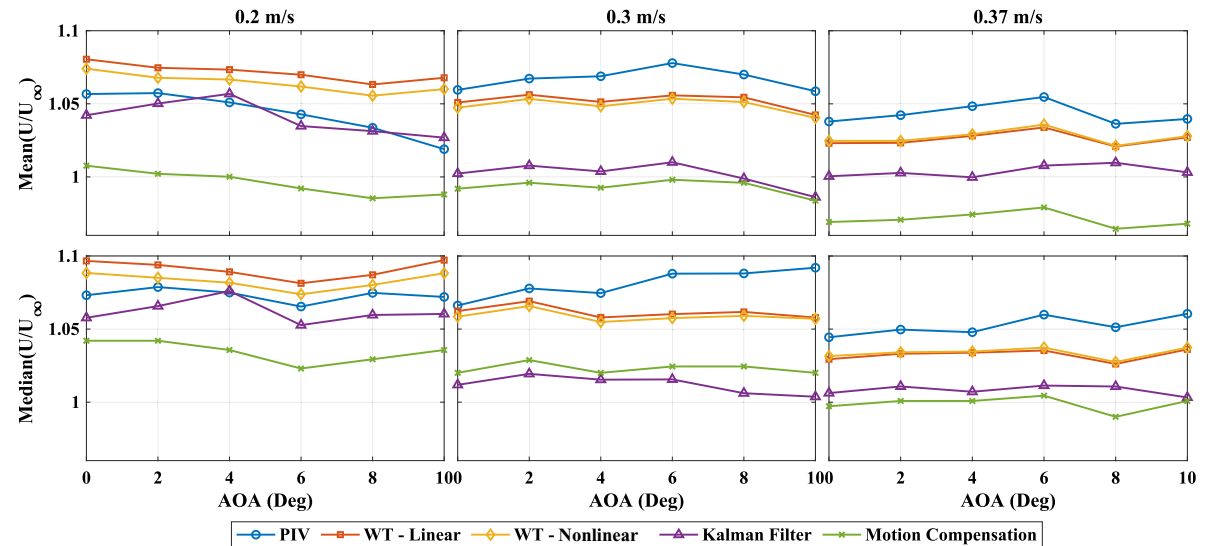


Fig. 20 Mean and median U-velocity versus angle of attack at three velocities; event-based methods replicate PIV trends, with WT methods showing closest agreement.

on a global scale, the event-based methods successfully replicate the trends observed in traditional PIV. However, consistent with the histograms in Fig. 19, a fixed bias error exists between PIV and the event-based methods across all cases. Among the event-based methods, the Weighted tracking methods (WT-Linear and WT-Non-Linear) show the closest agreement with PIV, while the Kalman Filter method follows. The Motion Compensation method exhibits the largest bias, highlighting its limitations in accurately capturing global velocity features.

Following the analysis of normalized U-velocity around the wing, a similar comparison was conducted for the normalized V-velocity. In most steady flows around wings, the V-velocity is typically an order of magnitude smaller than the U-velocity, making accurate estimation of V-velocity challenging, even for traditional PIV, due to limitations inherent in cross-correlation algorithms. Figure 21 shows the normalized V-velocity contours around the wing at $\alpha = 8^\circ$ for all three freestream velocities tested. Once again, the event-based methods demonstrate a reasonable match to the traditional PIV results, capturing key flow features around the wing. Because

of the high resolution achieved by tracking individual tracer particles, the event-based methods resolve flow features, particularly on the upper surface and in the wing's wake, more effectively than PIV. In contrast, the use of interrogation window sizes and 50% overlap in traditional PIV results in smoothed gradients, reducing its ability to resolve finer flow details in these regions.

The histograms corresponding to the contours in Fig. 21 are shown in Fig. 22 for different angles of attack α at a freestream velocity of 0.37 m/s. Interestingly, the constant bias observed in normalized U-velocity between PIV and the event-based methods is not present in the V-velocity results. Across all angles of attack, the event-based methods exhibit a higher degree of overlap with PIV, demonstrating their accuracy in capturing vertical velocity components. As the angle of attack increases, the V-velocity histograms broaden, similar to the trend observed for U-velocity, reflecting the increased variability in the flowfield caused by separation and stall. Notably, the peaks and valleys in the V-velocity histograms, associated with distinct regions of the flowfield (e.g., upper surface, wake), are accurately captured by the event-based methods. Among

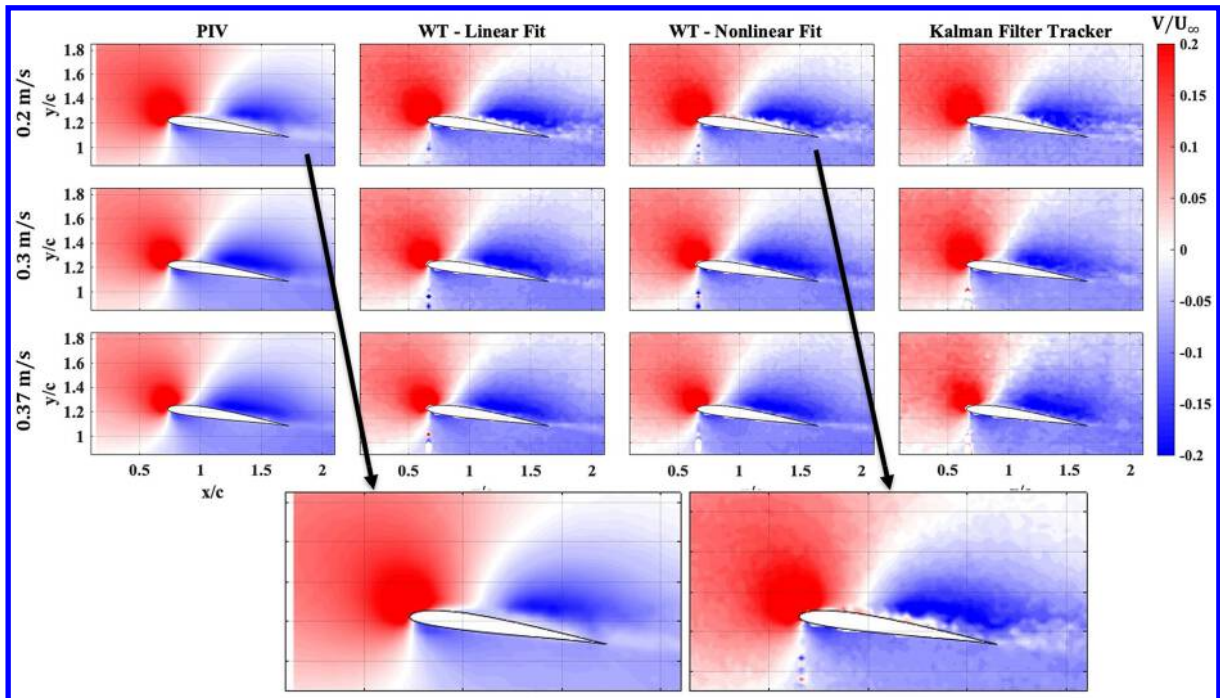


Fig. 21 Comparison of normalized V-velocity contours at $\alpha = 8^\circ$; event-based methods resolve upper surface and wake features with greater detail than PIV.

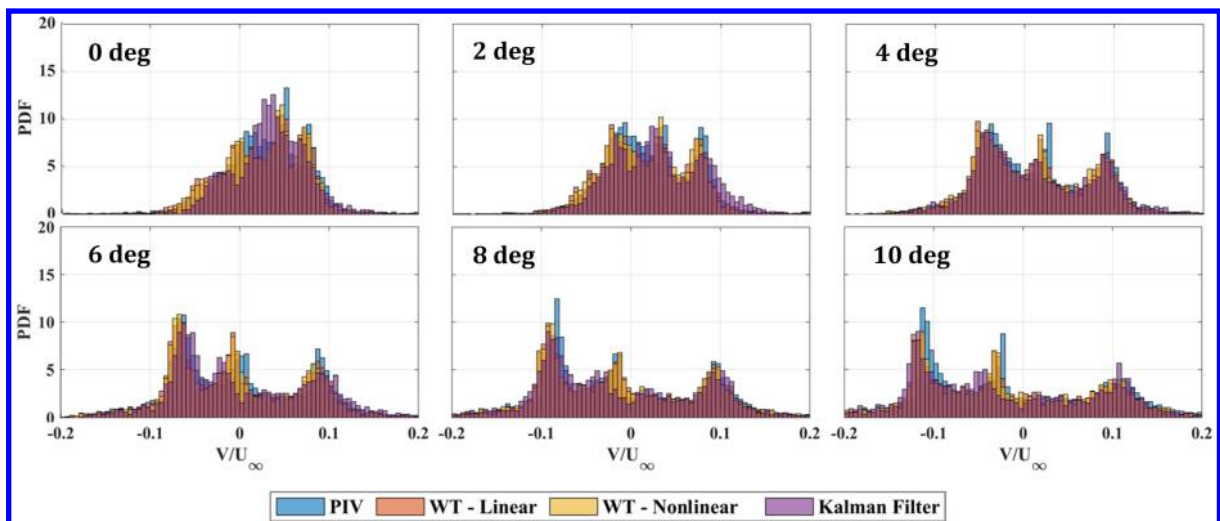


Fig. 22 PDFs (Probability Density Functions) of normalized V-velocity at $\alpha = 0$ to 10 deg and 0.37 m/s; event-based methods, especially WT, closely match PIV distributions.

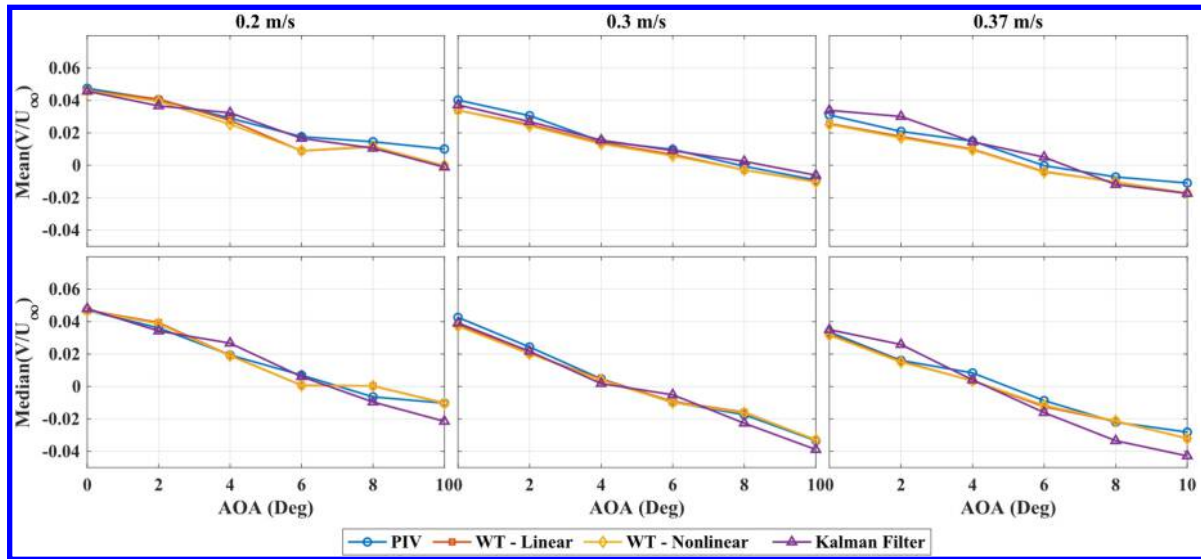


Fig. 23 Mean and median V-velocity versus angle of attack at three velocities; event-based methods match PIV trends and show no observable bias.

these, the Weighted tracking methods (WT-Linear and WT-Non-Linear) align most closely with PIV, while the Kalman Filter method exhibits greater deviation.

Figure 23 illustrates the variation of the mean and median of normalized V-velocity as a function of angle of attack α for all five methods and three freestream velocities tested. Similar to the results in Fig. 20, the mean and median values capture the V-velocity trends on the upper surface, lower surface, and wake, reflecting the features observed in Fig. 21. As the angle of attack increases, the mean and median V-velocity decrease, driven by increased flow separation and a corresponding rise in negative V-velocity components. The event-based methods closely replicate the trends and magnitudes of PIV, as shown in the histogram analysis in Fig. 22. However, it is unclear why a bias error is consistently observed in the normalized U-velocity estimates but not in the V-velocity estimates from the event-based methods.

Despite this uncertainty, the strong agreement between the mean, median, and overall distribution of U- and V-velocity from event-based methods and PIV highlights the suitability of event-based methods for velocimetry. These results reinforce their potential as a viable alternative to traditional PIV, particularly given their ability to capture flow features across the upper and lower surfaces and wake regions effectively.

V. Conclusions

This study presents a comprehensive evaluation of event-based methods for velocimetry in comparison to traditional PIV, focusing on the flow around an SD7003 wing at various freestream velocities and angles of attack. The results demonstrate that event-based methods, particularly the Weighted tracking algorithms (WT-Linear and WT-Non-Linear), offer strong agreement with PIV across a range of flow features, including velocity distributions, near-surface flow dynamics, and global velocity trends. Despite the inherent differences in data acquisition and processing approaches between event-based and frame-based techniques, the findings validate the capability of event-based methods to serve as reliable alternatives for flow diagnostics.

The analysis of normalized U and V-velocity contours and distributions highlighted several key findings:

- 1) Event-based methods effectively captured global trends in the mean and median of normalized U and V velocity across all tested velocities and angles of attack. Although a consistent bias error was observed in the U-velocity estimates, the V-velocity estimates showed no such bias, aligning closely with PIV results.

- 2) Event-based methods resolved near-surface flow features, such as displacement thickness and pressure gradients, more accurately than traditional PIV, which struggled with reflections and smoothing due to interrogation window overlap.

- 3) Event-based methods accurately captured the peaks and valleys in velocity histograms, reflecting the distinct regions of the flowfield. The Weighted tracking methods showed the closest agreement with PIV, while the Kalman Filter method exhibited some deviations.

- 4) Because of their particle-level resolution, event-based methods outperformed PIV in capturing sharp gradients and resolving subtle flow dynamics, particularly in regions of high velocity variability, such as the wing wake and areas of separation.

The findings also underscore the inherent advantages of event-based methods over PIV. Unlike PIV, which relies on optimized frame rates and interrogation window sizes that can smooth gradients and reduce resolution, event-based methods operate frame-free and track individual particles in real time. This allows them to provide high-resolution velocity data across all regions of the flowfield, including areas with steep gradients or low particle displacements.

The bias observed in the U-velocity is hypothesized to result from improper tuning of the methods. For example, the Kalman Filter approach involves several variables, such as measurement noise and initial conditions, that can significantly affect the accuracy of the final results. Similarly, the Motion Compensation algorithm relies on adjusting the observation frame to align with the expected flowfield velocity. The range and resolution of the expected flowfield velocity provided as input to this method can have a substantial impact on its performance. Even the Weighted tracking methods include several tuning parameters that can influence the accuracy of the final outcomes. Performing a comprehensive sensitivity analysis of these input parameters across all methods is the focus of future work and will be discussed in a follow-up paper.

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A. Broeren
Associate Editor